Contactor Control Test Report

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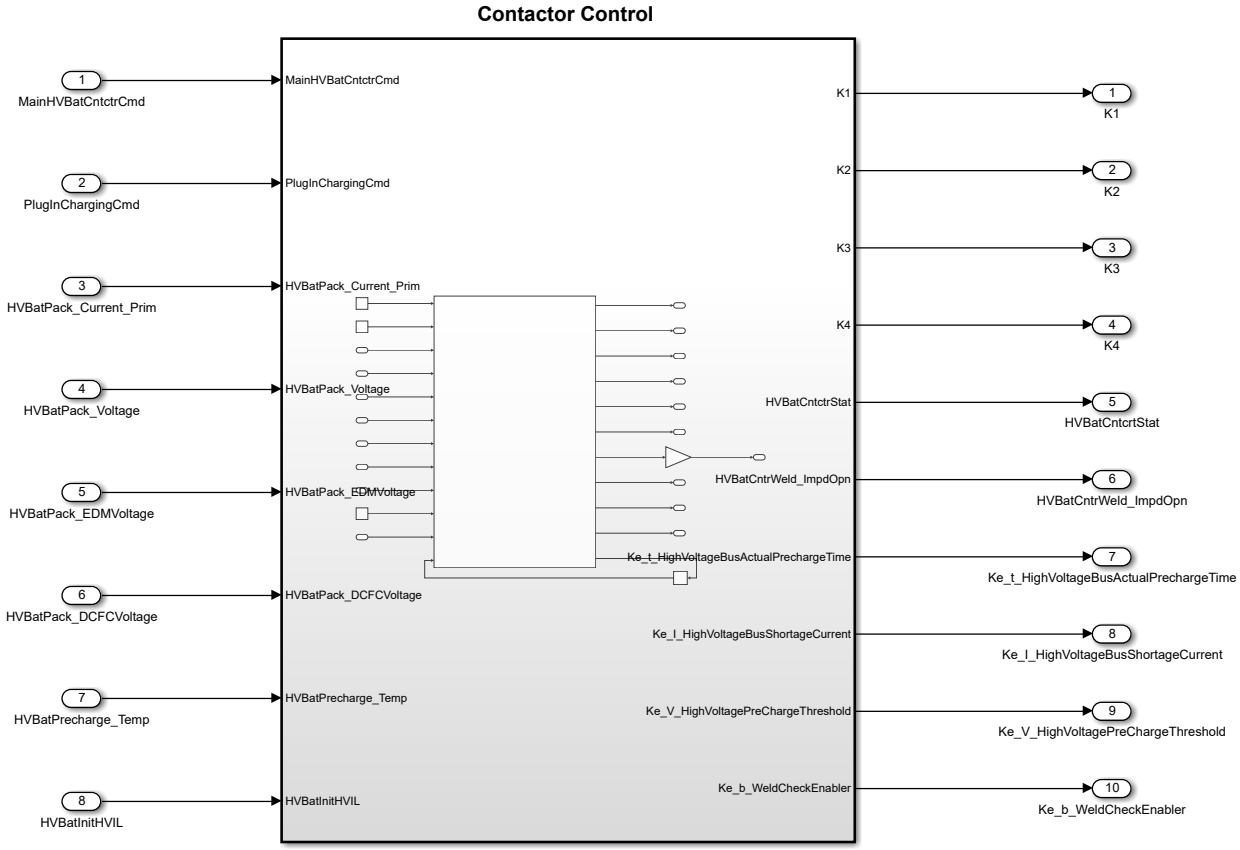
1. Contactor_Control_Test_Cases

1.1. Precharge_Test_Suite

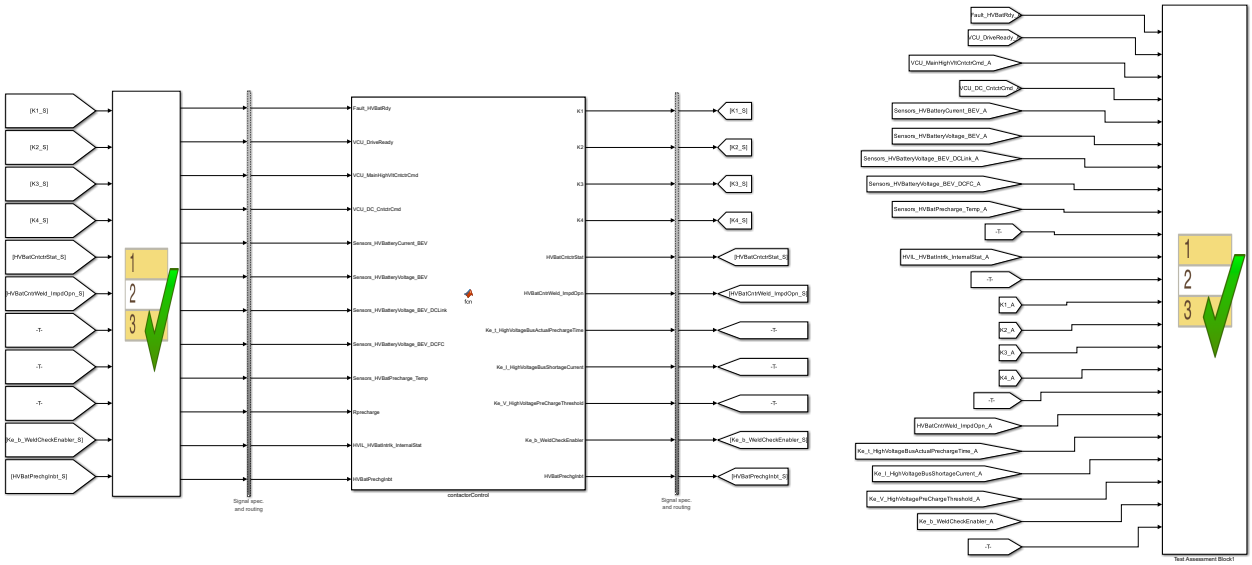
1.1.1. Precharge_Normal

System Under Test

Model Name: Contactor_Control_Standalone



Harness Name: Precharge_Normal



Test Sequence Data: Precharge_Normal/Test Assessment Block1

Symbols

Table 1.1. Input

Port	Name	Class	Data Type*	Size*
1	Fault_HVBatRdy	Data	double	[1]
2	VCU_DriveReady	Data	double	[1]
3	VCU_MainHighVltCntctrCmd	Data	double	[1]
4	VCU_DC_CntctrCmd	Data	double	[1]
5	Sensors_HVBatteryCurrent_BEV	Data	double	[1]
6	Sensors_HVBatteryVoltage_BEV	Data	double	[1]
7	Sensors_HVBatteryVoltage_BEV_DCLink	Data	double	[1]
8	Sensors_HVBatteryVoltage_BEV_DCFC	Data	double	[1]
9	Sensors_HVBatPrecharge_Temp	Data	double	[1]
10	Rprecharge	Data	double	[1]
11	HVIL_HVBatIntrlk_InternalStat	Data	double	[1]
12	HVBatPrechgInbt	Data	double	[1]
13	K1	Data	double	[1]
14	K2	Data	double	[1]
15	K3	Data	double	[1]
16	K4	Data	double	[1]
17	HVBatCntctrStat	Data	double	[1]

Port	Name	Class	Data Type*	Size*
18	HVBatCntrWeld_ImpdOpn	Data	double	[1]
19	Ke_t_HighVoltageBusActualPrechargeTime	Data	double	[1]
20	Ke_I_HighVoltageBusShortageCurrent	Data	double	[1]
21	Ke_V_HighVoltagePreChargeThreshold	Data	double	[1]
22	Ke_b_WeldCheckEnabler	Data	double	[1]
23	HVBatPrechgInbt1	Data	double	[1]

***Note:** The model was not compiled during report generation. As a result, the preceding Symbols section does not include information requiring model compilation. See the help for the TestSequence reporter for more information.*

Step Hierarchy

- step_1
  - step_1_1
  - Else
- step_2
  - step_2_1
  - Else
- step_3
  - step_3_1
  - Else
- step_4
  - step_4_1
  - Else
- step_5
  - step_5_1
  - Else
- step_5_1
  - step_5_1_1
  - Else
- step_6
  - step_6_1
  - Else
- step_7
  - step_7_1
  - Else
- step_8
  - step_8_1
  - Else

step_1

Table 1.2. Transition Table

Condition	Next Step
after(2,sec)	step_2

step_1_1

When Condition

```
true
```

Action

```
verify(K1 == uint64(0), '10.1.2.1.a')
```

Else

step_2

Table 1.3. Transition Table

Condition	Next Step
after(2,sec)	step_3

step_2_1

When Condition

```
true
```

Action

```
verify(K1 == uint64(0),'10.1.2.4')
```

Else

step_3

Table 1.4. Transition Table

Condition	Next Step
after(2,sec)	step_4

step_3_1

When Condition

```
true
```

Action

```
verify(K1 == uint64(0),'10.1.2.4, 10.1.2.5')
verify(K2 == uint64(0),'10.1.2.4')
verify(K3 == uint64(0),'10.1.2.4')
verify(K4 == uint64(0),'10.1.2.4')
```

Else

step_4

Table 1.5. Transition Table

Condition	Next Step
after(1002,msec)	step_5

step_4_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0),'10.1.2.6')
```

Else

step_5

Table 1.6. Transition Table

Condition	Next Step
after(1,msec)	step_5_1

step_5_1

Description

unit delay

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_5_1

Table 1.7. Transition Table

Condition	Next Step
after(92,msec)	step_6

step_5_1_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1),'10.2.0.1, 10.2.0.4,
10.2.0.5, 10.2.1.1')
verify(K1 == uint64(0),'10.2.0.1, 10.2.1.3')
verify(K2 == uint64(1),'10.2.0.1, 10.2.0.4, 10.2.0.5, 10.2.1.1,
10.2.1.2')
verify(K3 == uint64(1),'10.2.0.1')
verify(K4 == uint64(0),'10.2.0.1')
```

Else

step_6

Table 1.8. Transition Table

Condition	Next Step
after(1,msec)	step_7

step_6_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(2),'10.2.0.2, 10.2.0.3,
10.2.2.1, 10.2.2.4')
verify(K1 == uint64(1))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_7

Table 1.9. Transition Table

Condition	Next Step
after(1,sec)	step_8

step_7_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(2))
verify(K1 == uint64(1))
verify(K2 == uint64(0))
```

```
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_8

step_8_1

When Condition

```
true
```

Else

Test Sequence Data: Precharge_Normal/Test Sequence

Symbols

Table 1.10. Input

Port	Name	Class	Data Type*	Size*
1	K1	Data	double	[1]
2	K2	Data	double	[1]
3	K3	Data	double	[1]
4	K4	Data	double	[1]
5	HVBatCntctrStat	Data	double	[1]
6	HVBatCntrWeld_ImpdOpn	Data	double	[1]
7	Ke_t_HighVoltageBusActualPrechargeTime	Data	double	[1]
8	Ke_I_HighVoltageBusShortageCurrent	Data	double	[1]
9	Ke_V_HighVoltagePreChargeThreshold	Data	double	[1]
10	Ke_b_WeldCheckEnabler	Data	double	[1]
11	HVBatPrechgInbt	Data	double	[1]

Table 1.11. Output

Port	Name	Class	Data Type*	Size*
1	Fault_HVBatRdy	Data	double	[1]
2	VCU_DriveReady	Data	double	[1]
3	VCU_MainHighVltCntctrCmd	Data	double	[1]
4	VCU_DC_CntctrCmd	Data	double	[1]
5	Sensors_HVBatteryCurrent_BEV	Data	double	[1]

Port	Name	Class	Data Type*	Size*
6	Sensors_HVBatteryVoltage_BEV	Data	double	[1]
7	Sensors_HVBatteryVoltage_BEV_DCLink	Data	double	[1]
8	Sensors_HVBatteryVoltage_BEV_DCFC	Data	double	[1]
9	Sensors_HVBatPrecharge_Temp	Data	double	[1]
10	Rprecharge	Data	double	[1]
11	HVIL_HVBatIntrlk_InternalStat	Data	double	[1]
12	HVBatPrechgInbt1	Data	double	[1]

***Note:** The model was not compiled during report generation. As a result, the preceding Symbols section does not include information requiring model compilation. See the help for the TestSequence reporter for more information.*

Step Hierarchy

- Run
- step_1
- step_2
- step_3
- step_4
- step_5
- step_6
- step_7
- step_8

Run

Action

```
%% Initialize data outputs.
Fault_HVBatRdy = 0;
VCU_DriveReady = 0;
VCU_MainHighVltCntctrCmd = 0;
VCU_DC_CntctrCmd = 0;
Sensors_HVBatteryCurrent_BEV = 0;
Sensors_HVBatteryVoltage_BEV = 0;
Sensors_HVBatteryVoltage_BEV_DCLink = 0;
Sensors_HVBatteryVoltage_BEV_DCFC = 0;
Sensors_HVBatPrecharge_Temp = 0;
Rprecharge = 0;
HVIL_HVBatIntrlk_InternalStat = 0;
HVBatPrechgInbt1 = 0;
```

Table 1.12. Transition Table

Condition	Next Step
after(1,sec)	step_1

step_1

Action

```
VCU_MainHighVltCntctrCmd = 1;
```

Table 1.13. Transition Table

Condition	Next Step
after(2,sec)	step_2

step_2

Action

```
HVBatPrechgInbt1 = 0;
HVIL_HVBatIntrlk_InternalStat = 0;
Fault_HVBatRdy = 1;
Sensors_HVBatPrecharge_Temp= 25;
```

Table 1.14. Transition Table

Condition	Next Step
after(2,sec)	step_3

step_3

Description

MainHVBatCntctrCmd = 3 means fast_open. So, open all cntctrs

Action

```
VCU_MainHighVltCntctrCmd = 3;
```

Table 1.15. Transition Table

Condition	Next Step
after(2,sec)	step_4

step_4

Action

```
VCU_MainHighVltCntctrCmd = 1;
HVBatPrechgInbt1 = 0;
HVIL_HVBatIntrlk_InternalStat = 1;
Fault_HVBatRdy = 1;
Sensors_HVBatPrecharge_Temp= 25;
Rprecharge = 25;
Sensors_HVBatteryCurrent_BEV = 18;
Sensors_HVBatteryVoltage_BEV = 450;
Sensors_HVBatteryVoltage_BEV_DCLink = 2;
```

Table 1.16. Transition Table

Condition	Next Step
after(2,msec)	step_5

step_5

Description

precharge conditions are all satisfied

Action

```
Sensors_HVBatPrecharge_Temp= 26;
Sensors_HVBatteryCurrent_BEV = 1;
VCU_DriveReady = 1;
Sensors_HVBatteryVoltage_BEV = 450;
Sensors_HVBatteryVoltage_BEV_DCLink = 30;
```

Table 1.17. Transition Table

Condition	Next Step
after(3,msec)	step_6

step_6

Action

```
Sensors_HVBatPrecharge_Temp= 26;
Sensors_HVBatteryCurrent_BEV = 0.1;
Sensors_HVBatteryVoltage_BEV = 450;
Sensors_HVBatteryVoltage_BEV_DCLink = 60;
```

Table 1.18. Transition Table

Condition	Next Step
after(90,msec)	step_7

step_7

Action

```
Sensors_HVBatteryVoltage_BEV = 450;
Sensors_HVBatteryVoltage_BEV_DCLink = 450;
```

Table 1.19. Transition Table

Condition	Next Step
after(2,sec)	step_8

step_8

Simulation Settings Overrides

Simulation Mode	[Model Settings]
Releases	Current

Output Triggers

Start Logging	On simulation start
Stop Logging	When simulation stops
Shift time to zero	True

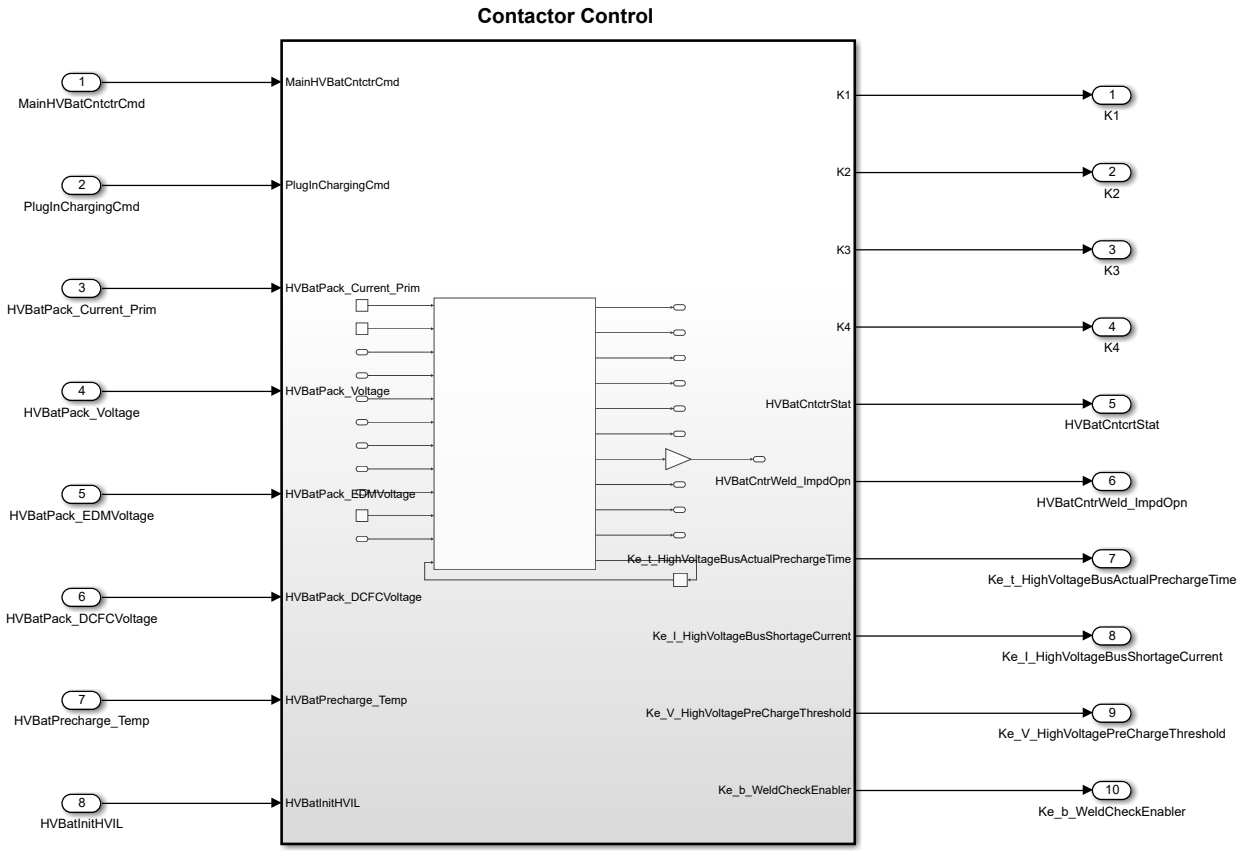
Configuration Settings Overrides

Configuration settings	Do not override model settings

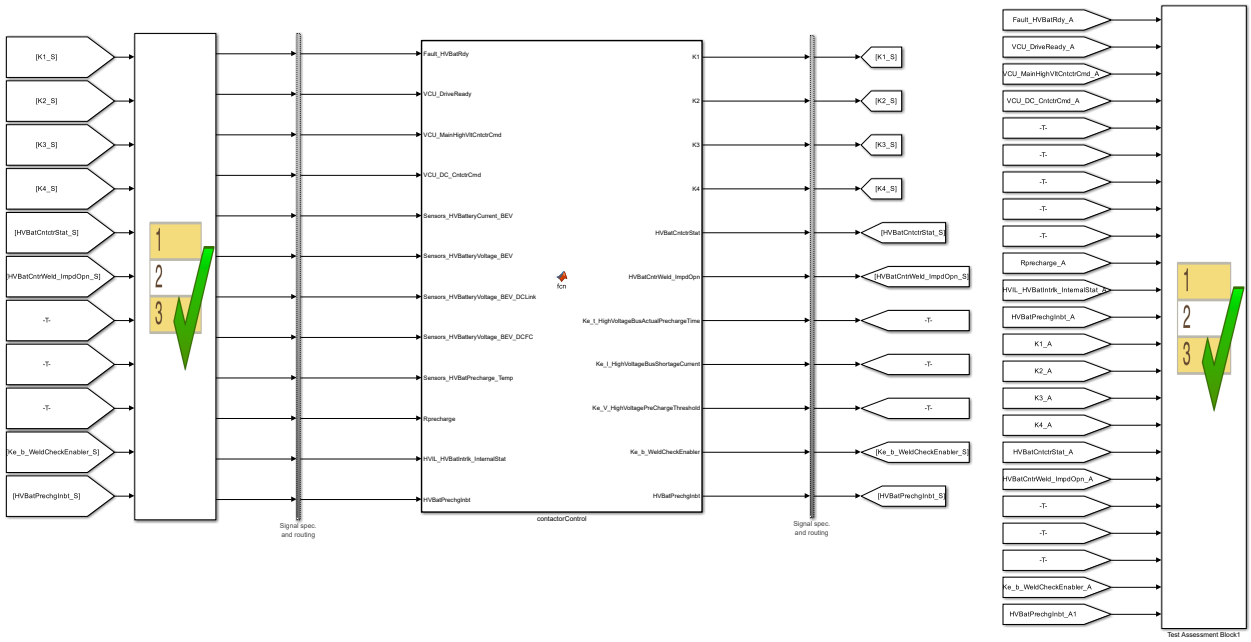
1.1.2. Precharge_Timeout

System Under Test

Model Name: Contactor_Control_Standalone



Harness Name: Precharge_Timeout



Test Sequence Data: Precharge_Timeout/Test Assessment Block1

Symbols

Table 1.20. Input

Port	Name	Class	Data Type*	Size*
1	Fault_HVBatRdy	Data	double	
2	VCU_DriveReady	Data	Inherit: Same as Simulink	-1
3	VCU_MainHighVltCntctrCmd	Data	double	[1]
4	VCU_DC_CntctrCmd	Data	double	[1]
5	Sensors_HVBatteryCurrent_BEV	Data	double	[1]
6	Sensors_HVBatteryVoltage_BEV	Data	double	[1]
7	Sensors_HVBatteryVoltage_BEV_DCLink	Data	double	[1]
8	Sensors_HVBatteryVoltage_BEV_DCFC	Data	double	[1]
9	Sensors_HVBatPrecharge_Temp	Data	double	[1]
10	Rprecharge	Data	double	[1]
11	HVIL_HVBatIntrlk_InternalStat	Data	double	[1]
12	HVBatPrechgInbt2	Data	double	[1]
13	K1	Data	double	[1]
14	K2	Data	double	[1]
15	K3	Data	double	[1]
16	K4	Data	double	[1]

Port	Name	Class	Data Type*	Size*
17	HVBatCntctrStat	Data	double	[1]
18	HVBatCntrWeld_ImpdOpn	Data	Inherit: Same as Simulink	-1
19	Ke_t_HighVoltageBusActualPrechargeTime	Data	double	[1]
20	Ke_I_HighVoltageBusShortageCurrent	Data	double	[1]
21	Ke_V_HighVoltagePreChargeThreshold	Data	double	[1]
22	Ke_b_WeldCheckEnabler	Data	Inherit: Same as Simulink	-1
23	HVBatPrechgInbt	Data	double	[1]

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Step Hierarchy

- step_1
  - step_1_1
  - Else
- step_2
  - step_2_1
  - Else
- step_3
  - step_3_1
  - Else
- step_4
  - step_4_1
  - Else
- step_5
  - step_5_1
  - Else
- step_6
  - step_6_1
  - Else
- step_7
  - step_7_1
  - Else
- step_8
  - step_8_1
  - Else
- step_9
  - step_9_1
  - Else
- step_10
  - step_10_1

    - Else
- step_11
    - step_11_1
    - Else
- step_12
    - step_12_1
    - Else
- step_13
    - step_13_1
    - Else
- step_14
    - step_14_1
    - Else
- step_15
    - step_15_1
    - Else
- step_16
    - step_16_1
    - Else
- step_17
    - step_17_1
    - Else
- step_18
    - step_18_1
    - Else
- step_19
    - step_19_1
    - Else
- step_20
    - step_20_1
    - Else
- step_21
    - step_21_1
    - Else
- step_22
    - step_22_1
    - Else
- step_23
    - step_23_1
    - Else
- step_24
    - step_24_1
    - Else
- step_25
    - step_25_1

    - Else
- step_26
    - step_26_1
    - Else
- step_27
    - step_27_1
    - Else
- step_28
    - step_28_1
    - Else
- step_29
    - step_29_1
    - Else
- step_30
    - step_30_1
    - Else
- step_31
    - step_31_1
    - Else
- step_32
    - step_32_1
    - Else
- step_33
    - step_33_1
    - Else
- step_34
    - step_34_1
    - Else
- step_35
    - step_35_1
    - Else
- step_36
    - step_36_1
    - Else
- step_37
    - step_37_1
    - Else
- step_38
    - step_38_1
    - Else
- step_39
    - step_39_1
    - Else
- step_40
    - step_40_1

    - Else
- step_41
    - step_41_1
    - Else
- step_42
    - step_42_1
    - Else
- step_43
    - step_43_1
    - Else
- step_44
    - step_44_1
    - Else
- step_45
    - step_45_1
    - Else
- step_46
    - step_46_1
    - Else
- step_47
    - step_47_1
    - Else
- step_48
    - step_48_1
    - Else
- step_49
    - step_49_1
    - Else
- step_50
    - step_50_1
    - Else
- step_51
    - step_51_1
    - Else
- step_52
    - step_52_1
    - Else
- step_53
    - step_53_1
    - Else
- step_54
    - step_54_1
    - Else
- step_55
    - step_55_1

    - Else
- step_56
    - step_56_1
    - Else

step_1

Table 1.21. Transition Table

Condition	Next Step
after(2,sec)	step_2

step_1_1

When Condition

```
true
```

Action

```
verify(K1 == uint64(0), '10.1.2.1.a')
```

Else

step_2

Table 1.22. Transition Table

Condition	Next Step
after(2,sec)	step_3

step_2_1

When Condition

```
true
```

Action

```
verify(K1 == uint64(0),'10.1.2.4')
```

Else

step_3

Table 1.23. Transition Table

Condition	Next Step
after(2,sec)	step_4

step_3_1

When Condition

```
true
```

Action

```
verify(K1 == uint64(0),'10.1.2.4')
verify(K2 == uint64(0),'10.1.2.4')
verify(K3 == uint64(0),'10.1.2.4')
verify(K4 == uint64(0),'10.1.2.4')
```

Else

step_4

Table 1.24. Transition Table

Condition	Next Step
after(1,sec)	step_5

step_4_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0),'10.1.2.6')
```

Else

step_5

Table 1.25. Transition Table

Condition	Next Step
after(1,msec)	step_6

step_5_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
```

Else

step_6

Table 1.26. Transition Table

Condition	Next Step
after(1,msec)	step_7

step_6_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_7

Table 1.27. Transition Table

Condition	Next Step
after(98,msec)	step_8

step_7_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_8

Table 1.28. Transition Table

Condition	Next Step
after(1,msec)	step_9

step_8_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_9

Table 1.29. Transition Table

Condition	Next Step
after(1001,msec)	step_10

step_9_1

Description

first timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3),'10.2.3.1, 10.2.3.10')
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_10

Table 1.30. Transition Table

Condition	Next Step
after(1,msec)	step_11

step_10_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_11

Table 1.31. Transition Table

Condition	Next Step
after(1,msec)	step_12

step_11_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1),'10.2.3.2')
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_12

Table 1.32. Transition Table

Condition	Next Step
after(100,msec)	step_13

step_12_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1),'10.2.3.5, 10.2.3.6')
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_13

Table 1.33. Transition Table

Condition	Next Step
after(1001,msec)	step_14

step_13_1

Description

second timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_14

Table 1.34. Transition Table

Condition	Next Step
after(1,msec)	step_15

step_14_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_15

Table 1.35. Transition Table

Condition	Next Step
after(1,msec)	step_16

step_15_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_16

Table 1.36. Transition Table

Condition	Next Step
after(100,msec)	step_17

step_16_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_17

Table 1.37. Transition Table

Condition	Next Step
after(1001,msec)	step_18

step_17_1

Description

third timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_18

Table 1.38. Transition Table

Condition	Next Step
after(1,msec)	step_19

step_18_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_19

Table 1.39. Transition Table

Condition	Next Step
after(1,msec)	step_20

step_19_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_20

Table 1.40. Transition Table

Condition	Next Step
after(100,msec)	step_21

step_20_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_21

Table 1.41. Transition Table

Condition	Next Step
after(1001,msec)	step_22

step_21_1

Description

fourth timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_22

Table 1.42. Transition Table

Condition	Next Step
after(1,msec)	step_23

step_22_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_23

Table 1.43. Transition Table

Condition	Next Step
after(1,msec)	step_24

step_23_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_24

Table 1.44. Transition Table

Condition	Next Step
after(100,msec)	step_25

step_24_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_25

Table 1.45. Transition Table

Condition	Next Step
after(1001,msec)	step_26

step_25_1

Description

fifth timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_26

Table 1.46. Transition Table

Condition	Next Step
after(1,msec)	step_27

step_26_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_27

Table 1.47. Transition Table

Condition	Next Step
after(1,msec)	step_28

step_27_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_28

Table 1.48. Transition Table

Condition	Next Step
after(100,msec)	step_29

step_28_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_29

Table 1.49. Transition Table

Condition	Next Step
after(1001,msec)	step_30

step_29_1

Description

sixth timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_30

Table 1.50. Transition Table

Condition	Next Step
after(1,msec)	step_31

step_30_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_31

Table 1.51. Transition Table

Condition	Next Step
after(1,msec)	step_32

step_31_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_32

Table 1.52. Transition Table

Condition	Next Step
after(100,msec)	step_33

step_32_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_33

Table 1.53. Transition Table

Condition	Next Step
after(1001,msec)	step_34

step_33_1

Description

seventh timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_34

Table 1.54. Transition Table

Condition	Next Step
after(1,msec)	step_35

step_34_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_35

Table 1.55. Transition Table

Condition	Next Step
after(1,msec)	step_36

step_35_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_36

Table 1.56. Transition Table

Condition	Next Step
after(100,msec)	step_37

step_36_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_37

Table 1.57. Transition Table

Condition	Next Step
after(1001,msec)	step_38

step_37_1

Description

eighth timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_38

Table 1.58. Transition Table

Condition	Next Step
after(1,msec)	step_39

step_38_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_39

Table 1.59. Transition Table

Condition	Next Step
after(1,msec)	step_40

step_39_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_40

Table 1.60. Transition Table

Condition	Next Step
after(100,msec)	step_41

step_40_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_41

Table 1.61. Transition Table

Condition	Next Step
after(1001,msec)	step_42

step_41_1

Description

ninth timeout

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_42

Table 1.62. Transition Table

Condition	Next Step
after(1,msec)	step_43

step_42_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_43

Table 1.63. Transition Table

Condition	Next Step
after(1,msec)	step_44

step_43_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_44

Table 1.64. Transition Table

Condition	Next Step
after(100,msec)	step_45

step_44_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_45

Table 1.65. Transition Table

Condition	Next Step
after(1001,msec)	step_46

step_45_1

Description

tenth timeout becomes precharge inhibit

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(4))
verify(K1 == uint64(0))
verify(K2 == uint64(0),'10.2.3.7, 10.2.3.8, 10.2.5.1')
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_46

Table 1.66. Transition Table

Condition	Next Step
after(14000,msec)	step_47

step_46_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(4),'10.2.3.9')
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_47

Table 1.67. Transition Table

Condition	Next Step
after(1,msec)	step_48

step_47_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_48

Table 1.68. Transition Table

Condition	Next Step
after(1,msec)	step_49

step_48_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_49

Table 1.69. Transition Table

Condition	Next Step
after(100,msec)	step_50

step_49_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_50

Table 1.70. Transition Table

Condition	Next Step
after(1001,msec)	step_51

step_50_1

Description

first timeout in the new cycle of reattempt

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(3))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_51

Table 1.71. Transition Table

Condition	Next Step
after(1,msec)	step_52

step_51_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_52

Table 1.72. Transition Table

Condition	Next Step
after(1967,msec)	step_53

step_52_1

Description

Precharge resistor temperature is too high set by test sequence

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(0),'10.2.5.2')
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_53

Table 1.73. Transition Table

Condition	Next Step
after(1,msec)	step_54

step_53_1

Description

Precharge resistor temperature is normal set by test sequence. So, precharging starts again.

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_54

Table 1.74. Transition Table

Condition	Next Step
after(10,msec)	step_55

step_54_1

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(1))
verify(K1 == uint64(0))
verify(K2 == uint64(1))
verify(K3 == uint64(1))
verify(K4 == uint64(0))
```

Else

step_55

step_55_1

Description

precharge criteria are finally met (Successful precharge after multiple previous timeout)

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == uint64(2))
verify(K1 == uint64(1))
verify(K2 == uint64(0))
verify(K3 == uint64(0))
verify(K4 == uint64(0))
```

Else

step_56

step_56_1

When Condition

```
true
```

Else

Test Sequence Data: Precharge_Timeout/Test Sequence

Symbols

Table 1.75. Input

Port	Name	Class	Data Type*	Size*
1	K1	Data	double	[1]
2	K2	Data	double	[1]
3	K3	Data	double	[1]
4	K4	Data	double	[1]
5	HVBatCntctrStat	Data	double	[1]
6	HVBatCntrWeld_ImpdOpn	Data	double	[1]
7	Ke_t_HighVoltageBusActualPrechargeTime	Data	double	[1]
8	Ke_I_HighVoltageBusShortageCurrent	Data	double	[1]
9	Ke_V_HighVoltagePreChargeThreshold	Data	double	[1]
10	Ke_b_WeldCheckEnabler	Data	double	[1]
11	HVBatPrechgInbt	Data	double	[1]

Table 1.76. Output

Port	Name	Class	Data Type*	Size*
1	Fault_HVBatRdy	Data	double	[1]
2	VCU_DriveReady	Data	double	[1]
3	VCU_MainHighVltCntctrCmd	Data	double	[1]
4	VCU_DC_CntctrCmd	Data	double	[1]
5	Sensors_HVBatteryCurrent_BEV	Data	double	[1]
6	Sensors_HVBatteryVoltage_BEV	Data	double	[1]
7	Sensors_HVBatteryVoltage_BEV_DCLink	Data	double	[1]
8	Sensors_HVBatteryVoltage_BEV_DCFC	Data	double	[1]
9	Sensors_HVBatPrecharge_Temp	Data	double	[1]
10	Rprecharge	Data	double	[1]
11	HVIL_HVBatIntrlk_InternalStat	Data	double	[1]
12	HVBatPrechgInbt1	Data	double	[1]

***Note:** The model was not compiled during report generation. As a result, the preceding Symbols section does not include information requiring model compilation. See the help for the TestSequence reporter for more information.*

Step Hierarchy

- Run
- step_1
- step_2
- step_3
- step_4
- step_5
- step_6
- step_7
- step_8
- step_9
- step_10

Run

Action

```
%% Initialize data outputs.
Fault_HVBatRdy = 0;
VCU_DriveReady = 0;
VCU_MainHighVltCntctrCmd = 0;
VCU_DC_CntctrCmd = 0;
Sensors_HVBatteryCurrent_BEV = 0;
Sensors_HVBatteryVoltage_BEV = 0;
Sensors_HVBatteryVoltage_BEV_DCLink = 0;
Sensors_HVBatteryVoltage_BEV_DCFC = 0;
Sensors_HVBatPrecharge_Temp = 0;
Rprecharge = 25;
HVIL_HVBatIntrlk_InternalStat = 0;
HVBatPrechgInbt1 = 0;
```

Table 1.77. Transition Table

Condition	Next Step
after(1,sec)	step_1

step_1

Action

```
VCU_MainHighVltCntctrCmd  = 1;
```

Table 1.78. Transition Table

Condition	Next Step
after(2,sec)	step_2

step_2

Action

```
HVBatPrechgInbt1 = 0;
HVIL_HVBatIntrlk_InternalStat = 0;
Fault_HVBatRdy = 1;
Sensors_HVBatPrecharge_Temp = 25;
```

Table 1.79. Transition Table

Condition	Next Step
after(2,sec)	step_3

step_3

Action

```
VCU_MainHighVltCntctrCmd  = 3;
```

Table 1.80. Transition Table

Condition	Next Step
after(2,sec)	step_4

step_4

Action

```
VCU_MainHighVltCntctrCmd  = 1;
HVBatPrechgInbt1 = 0;
HVIL_HVBatIntrlk_InternalStat = 1;
Fault_HVBatRdy = 1;
Sensors_HVBatPrecharge_Temp = 25;
Rprecharge = 25;
Sensors_HVBatteryCurrent_BEV = 18;
VCU_DriveReady = 1;
Sensors_HVBatteryVoltage_BEV = 450;
```

Table 1.81. Transition Table

Condition	Next Step
after(2,msec)	step_5

step_5

Action

```
Sensors_HVBatPrecharge_Temp = 26;
Sensors_HVBatteryCurrent_BEV = 1;
Sensors_HVBatteryVoltage_BEV_DCLink = 50;
```

Table 1.82. Transition Table

Condition	Next Step
after(3,msec)	step_6

step_6

Action

```
Sensors_HVBatPrecharge_Temp = 26;
Sensors_HVBatteryCurrent_BEV = 0.1;
Sensors_HVBatteryVoltage_BEV_DCLink = 350;
```

Table 1.83. Transition Table

Condition	Next Step
after(95,msec)	step_7

step_7

Action

```
Sensors_HVBatteryVoltage_BEV_DCLink = 350;
```

Table 1.84. Transition Table

Condition	Next Step
after(26,sec)	step_8

step_8

Action

```
Sensors_HVBatPrecharge_Temp = 61;
```

Table 1.85. Transition Table

Condition	Next Step
after(2,sec)	step_9

step_9

Action

```
Sensors_HVBatPrecharge_Temp = 50;
```

Table 1.86. Transition Table

Condition	Next Step
after(11,msec)	step_10

step_10

Action

```
Sensors_HVBatteryVoltage_BEV = 450;
Sensors_HVBatteryVoltage_BEV_DCLink = 450;
```

Simulation Settings Overrides

Simulation Mode	[Model Settings]
Releases	Current

Output Triggers

Start Logging	On simulation start
Stop Logging	When simulation stops

Shift time to zero	True

Configuration Settings Overrides

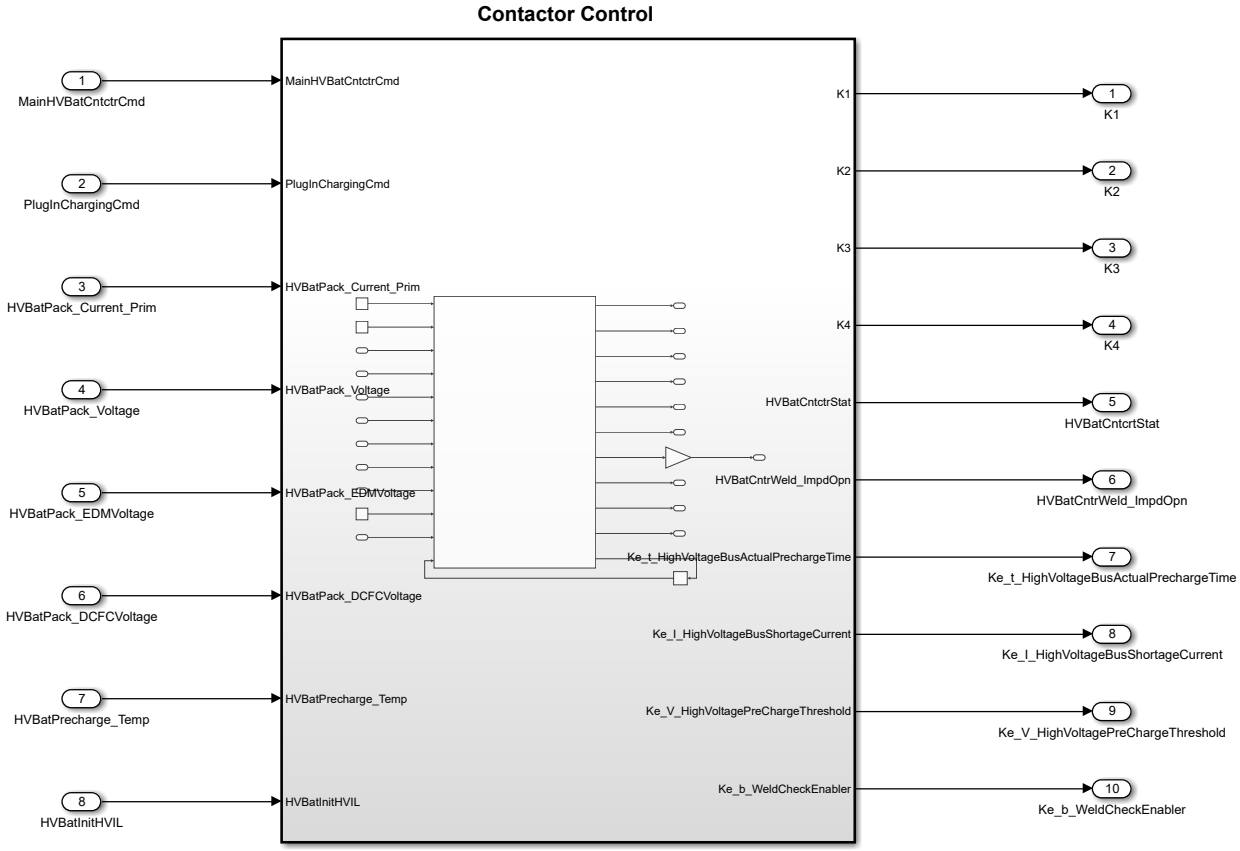
Configuration settings	Do not override model settings

1.2. Weld_Check_Diagnostics

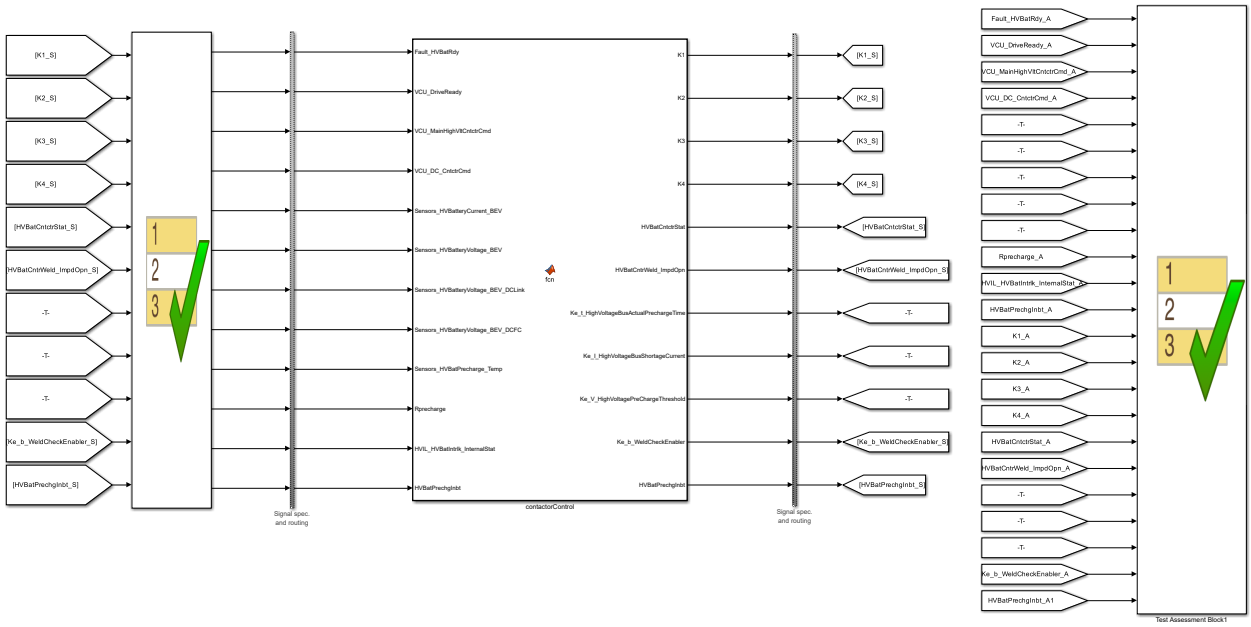
1.2.1. Stuck_Open_or_Closed

System Under Test

Model Name: Contactor_Control_Standalone



Harness Name: Weld_Check_Diagnostics



Test Sequence Data: Weld_Check_Diagnostics/Test Assessment Block1

Symbols

Table 1.87. Input

Port	Name	Class	Data Type*	Size*
1	Fault_HVBatRdy	Data	double	[1]
2	VCU_DriveReady	Data	double	[1]
3	VCU_MainHighVltCntctrCmd	Data	double	[1]
4	VCU_DC_CntctrCmd	Data	double	[1]
5	Sensors_HVBatteryCurrent_BEV	Data	double	[1]
6	Sensors_HVBatteryVoltage_BEV	Data	double	[1]
7	Sensors_HVBatteryVoltage_BEV_DCLink	Data	double	[1]
8	Sensors_HVBatteryVoltage_BEV_DCFC	Data	double	[1]
9	Sensors_HVBatPrecharge_Temp	Data	double	[1]
10	Rprecharge	Data	double	[1]
11	HVIL_HVBatIntrlk_InternalStat	Data	double	[1]
12	HVBatPrechgInbt	Data	double	[1]
13	K1	Data	double	[1]
14	K2	Data	double	[1]
15	K3	Data	double	[1]
16	K4	Data	double	[1]

Port	Name	Class	Data Type*	Size*
17	HVBatCntctrStat	Data	double	[1]
18	HVBatCntrWeld_ImpdOpn	Data	Inherit: Same as Simulink	-1
19	Ke_t_HighVoltageBusActualPrechargeTime	Data	double	[1]
20	Ke_I_HighVoltageBusShortageCurrent	Data	double	[1]
21	Ke_V_HighVoltagePreChargeThreshold	Data	double	[1]
22	Ke_b_WeldCheckEnabler	Data	Inherit: Same as Simulink	-1
23	HVBatPrechgInbt1	Data	double	[1]

***Note:** The model was not compiled during report generation. As a result, the preceding Symbols section does not include information requiring model compilation. See the help for the TestSequence reporter for more information.*

Step Hierarchy

- Run
  - step_1_1
  - Else
- step_1
  - step_1_1
  - Else
- step_2
  - step_2_1
  - Else
- step_3
  - step_3_1
  - Else
- step_4
  - step_4_1
  - Else
- step_5
  - step_5_1
  - Else
- step_6
  - step_6_1
  - Else
- step_7
  - step_7_1
  - Else
- step_8
  - step_8_1
  - Else

Run

Table 1.88. Transition Table

Condition	Next Step
after(1,sec)	step_1

step_1_1

Description

open

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == 0);
verify(HVBatCntrWeld_ImpdOpn == 0);
verify(Ke_b_WeldCheckEnabler == 1);
```

Else

step_1

Table 1.89. Transition Table

Condition	Next Step
after(1,sec)	step_2

step_1_1

Description

Shorted High (stuck closed)

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == 6,'10.3.0.2, 10.3.0.3.a, 10.1.4.2');
verify(K1 == 0);
```

```
verify(HVBatCntrWeld_ImpdOpn == 2,'10.3.0.2');
verify(Ke_b_WeldCheckEnabler == 1);
```

Else

step_2

Table 1.90. Transition Table

Condition	Next Step
after(1,sec)	step_3

step_2_1

Description

open

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == 0,'10.3.0.1.a, 10.3.0.1.b');
verify(HVBatCntrWeld_ImpdOpn == 0,'10.3.0.2.a');
verify(Ke_b_WeldCheckEnabler == 1);
```

Else

step_3

Table 1.91. Transition Table

Condition	Next Step
after(1,msec)	step_4

step_3_1

Description

Precharging

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == 1);
verify(Ke_b_WeldCheckEnabler == 1);
verify(HVBatCntrWeld_ImpdOpn == 0);
```

Else

step_4

Table 1.92. Transition Table

Condition	Next Step
after(999,msec)	step_5

step_4_1

Description

main normal closed

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == 2);
verify(Ke_b_WeldCheckEnabler == 1);
verify(HVBatCntrWeld_ImpdOpn == 0);
```

Else

step_5

Table 1.93. Transition Table

Condition	Next Step
after(1,sec)	step_6

step_5_1

Description

shorted low (stuck open)

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == 7, '10.3.0.3.a, 10.1.4.1, 10.1.4.2,
10.1.4.3');
verify(Ke_b_WeldCheckEnabler == 1);
verify(HVBatCntrWeld_ImpdOpn == 1,'10.1.4.4');
```

Else

step_6

Table 1.94. Transition Table

Condition	Next Step
after(1,sec)	step_7

step_6_1

Description

main normal closed

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == 2);
verify(Ke_b_WeldCheckEnabler == 1);
verify(HVBatCntrWeld_ImpdOpn == 0,'10.1.4.4');
```

Else

step_7

Table 1.95. Transition Table

Condition	Next Step
after(1,sec)	step_8

step_7_1

Description

impact open (weld check diagnostic disabled)

When Condition

```
true
```

Action

```
verify(HVBatCntctrStat == 0);
verify(Ke_b_WeldCheckEnabler == 0,'10.3.1.2.c, 10.3.1.3');
verify(HVBatCntrWeld_ImpdOpn == 0);
```

Else

step_8

step_8_1

When Condition

```
true
```

Else

Test Sequence Data: Weld_Check_Diagnostics/Test Sequence

Symbols

Table 1.96. Input

Port	Name	Class	Data Type*	Size*
1	K1	Data	double	[1]
2	K2	Data	double	[1]
3	K3	Data	double	[1]
4	K4	Data	double	[1]
5	HVBatCntctrStat	Data	double	[1]
6	HVBatCntrWeld_ImpdOpn	Data	double	[1]
7	Ke_t_HighVoltageBusActualPrechargeTime	Data	double	[1]
8	Ke_I_HighVoltageBusShortageCurrent	Data	double	[1]
9	Ke_V_HighVoltagePreChargeThreshold	Data	double	[1]

Port	Name	Class	Data Type*	Size*
10	Ke_b_WeldCheckEnabler	Data	double	[1]
11	HVBatPrechgInbt	Data	double	[1]

Table 1.97. Output

Port	Name	Class	Data Type*	Size*
1	Fault_HVBatRdy	Data	double	[1]
2	VCU_DriveReady	Data	double	[1]
3	VCU_MainHighVltCntctrCmd	Data	double	[1]
4	VCU_DC_CntctrCmd	Data	double	[1]
5	Sensors_HVBatteryCurrent_BEV	Data	double	[1]
6	Sensors_HVBatteryVoltage_BEV	Data	double	[1]
7	Sensors_HVBatteryVoltage_BEV_DCLink	Data	double	[1]
8	Sensors_HVBatteryVoltage_BEV_DCFC	Data	double	[1]
9	Sensors_HVBatPrecharge_Temp	Data	double	[1]
10	Rprecharge	Data	double	[1]
11	HVIL_HVBatIntrlk_InternalStat	Data	double	[1]
12	HVBatPrechgInbt1	Data	double	[1]

***Note:** The model was not compiled during report generation. As a result, the preceding Symbols section does not include information requiring model compilation. See the help for the TestSequence reporter for more information.*

Step Hierarchy

- Run
- step_1
- step_2
- step_3
- step_4
- step_5
- step_6
- step_7

Run

Action

```
%% Initialize data outputs.
Fault_HVBatRdy = 1;
VCU_DriveReady = 1;
VCU_MainHighVltCntctrCmd = 0;
```

```
VCU_DC_CntctrCmd = 0;
Sensors_HVBatteryCurrent_BEV = 0;
Sensors_HVBatteryVoltage_BEV = 435;
Sensors_HVBatteryVoltage_BEV_DCLink = 0;
Sensors_HVBatteryVoltage_BEV_DCFC = 0;
Sensors_HVBatPrecharge_Temp = 0;
Rprecharge = 25;
HVIL_HVBatIntrlk_InternalStat = 1;
HVBatPrechgInbt1 = 0;
```

Table 1.98. Transition Table

Condition	Next Step
after(1,sec)	step_1

step_1

Action

```
VCU_MainHighVltCntctrCmd = 0;
Sensors_HVBatteryVoltage_BEV_DCLink = 272;
```

Table 1.99. Transition Table

Condition	Next Step
after(1,sec)	step_2

step_2

Action

```
VCU_MainHighVltCntctrCmd = 0;
Sensors_HVBatteryVoltage_BEV_DCLink = 268;
```

Table 1.100. Transition Table

Condition	Next Step
after(1,sec)	step_3

step_3

Action

```
VCU_MainHighVltCntctrCmd = 1;
Sensors_HVBatteryVoltage_BEV_DCLink = 430;
Sensors_HVBatteryCurrent_BEV = 0.5;
```

Table 1.101. Transition Table

Condition	Next Step
after(1,sec)	step_4

step_4

Action

```
VCU_MainHighVltCntctrCmd = 1;
Sensors_HVBatteryVoltage_BEV_DCLink = 269;
```

Table 1.102. Transition Table

Condition	Next Step
after(1,sec)	step_5

step_5

Action

```
VCU_MainHighVltCntctrCmd = 1;
Sensors_HVBatteryVoltage_BEV_DCLink = 271;
```

Table 1.103. Transition Table

Condition	Next Step
after(1,sec)	step_6

step_6

Action

```
VCU_MainHighVltCntctrCmd = 2;
```

Table 1.104. Transition Table

Condition	Next Step
after(1,sec)	step_7

step_7

Simulation Settings Overrides

Simulation Mode	[Model Settings]
Releases	Current

Output Triggers

Start Logging	On simulation start
Stop Logging	When simulation stops
Shift time to zero	True

Configuration Settings Overrides

Configuration settings	Do not override model settings